$$B(\rho, c) = \frac{\rho^c / c!}{\sum_{j=0}^{c} \frac{\rho^j}{j!}} \quad \text{—— (A)}$$

The recursive form

$$B(\rho, c) = \frac{\rho\, B(\rho, c-1)}{c + \rho\, B(\rho, c-1)} \quad \text{—— (B)}$$

we can write (A) as

$$B(\rho, c) = \frac{\frac{\rho}{c}\left[\frac{\rho^{(c-1)}}{(c-1)!}\right] \times \frac{1}{\sum_{j=0}^{c-1} \frac{\rho^j}{j!}}}{\frac{\rho^c}{c!} + \sum_{j=0}^{c-1} \frac{\rho^j}{j!} \times \frac{1}{\sum_{j=0}^{c-1} \frac{\rho^j}{j!}}}$$

$$= \frac{\rho}{c} \cdot \underbrace{\frac{\rho^{c-1}/(c-1)!}{\sum_{j=0}^{c-1} \rho^j / j}}_{B(\rho, c-1)}$$

$$B(\rho, c-1) \cdot \frac{\times}{\frac{\rho}{c}\left[\frac{\rho^{c-1}}{(c-1)!} \Big/ \sum_{j=0}^{c-1} \frac{\rho^j}{j!}\right] + \underbrace{\left\{\frac{\sum_{j=0}^{c-1} \rho^j / j!}{\sum_{j=0}^{c-1} \rho^j / j!}\right\}}_{1} \times c} \cdot \frac{c}{c}$$

$$= \frac{\rho\, B(\rho, c-1)}{B(\rho, c-1) + c} \Longrightarrow \text{(B)}$$